

Integrated Lifecare Platform



Nurse Call System Analytics, Nurse Call Administration, Mobile apps
and Real Time Healthcare System

Overview

In healthcare and life sciences, Honeywell offers a comprehensive suite of solutions designed to enhance patient safety, improve clinical outcomes, and boost caregiver productivity. The UL 1069-certified Systevo Nurse Call system forms the backbone of patient-caregiver communication, integrating seamlessly with mobile applications like Systevo Connected Life Care to reduce response times and alarm fatigue. This is augmented by the Real Time Healthcare System (RTHS) which uses wireless body patches for real-time, continuous vital signs monitoring both in-hospital and remotely. Further solutions include the Integrated LifeCare (ILC) Medication App for safe and compliant medication administration, Operational Intelligence software for optimizing clinical mobility devices, and strategic partnerships, such as with CenTrak for Real-Time Location Systems (RTLS), to create a fully connected and efficient hospital ecosystem.

The modern hospital environment is characterized by intense pressure and disjointed digital tools, leading to workflow friction for nursing staff. Historically, critical activities like patient requests (nurse calls), medication administration, and vital sign documentation relied on separate, often cumbersome systems. Nurses were performing 15–25 scans per shift while simultaneously balancing multiple devices and urgent tasks. The fragmented user experience in the existing applications resulted in confusion, potential delays in patient response, and inconsistency in logging critical data.

The Problem

Cognitive Overload in Critical Care

Healthcare professionals were overwhelmed by unorganized raw data regarding nurse calls and alerts. Without a centralized way to visualize ward activity, they struggled to quickly interpret complex metrics—such as active calls and ward workload—leading to increased cognitive strain and slower decision-making during critical moments.

How might we transform complex, real-time hospital data into a digestible, visual dashboard that allows staff to prioritize tasks instantly without mental fatigue?

Traditional nurse call systems and older visualizations often contributed to alarm fatigue due to poor information hierarchy. For instance, looking at the previous dashboard designs, the interpretation of alert

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system into immediate, intelligent, and actionable alerts for mobile caregivers.

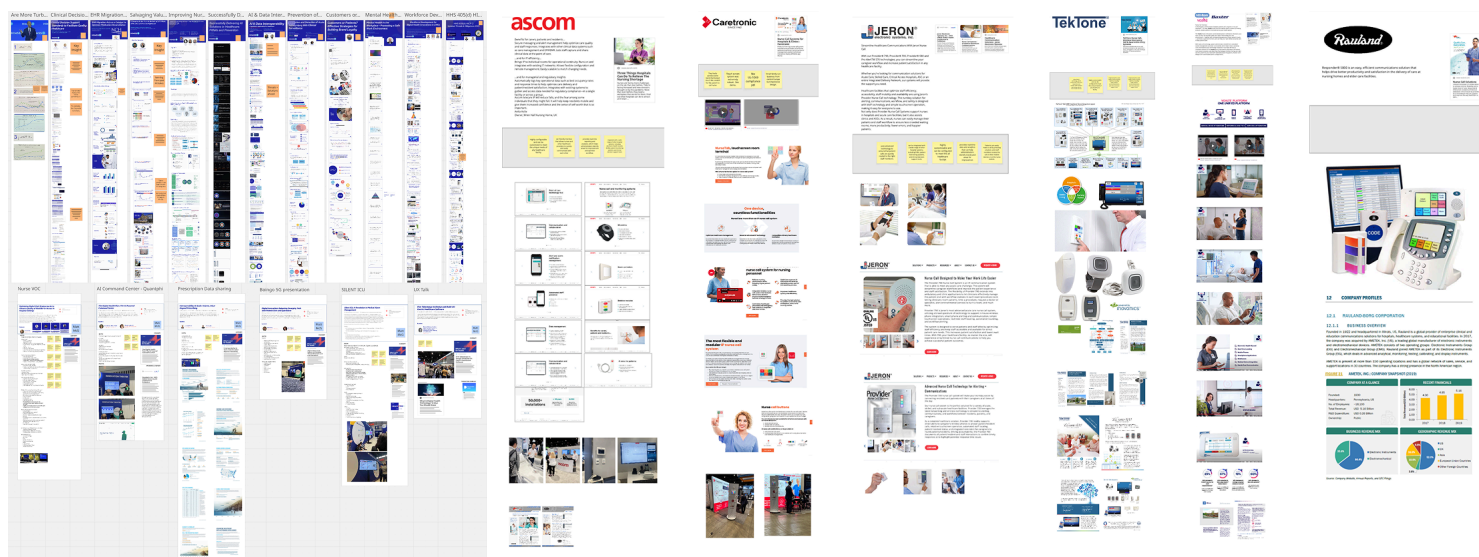
The Goal

The primary goal of the Integrated Lifecare (ILC) Platform initiative was to unify these fragmented clinical workflows and communication channels into a single, intuitive mobile-first platform. This unification aimed to increase caregiver efficiency, reduce cognitive load, decrease administrative task time, and ultimately improve patient safety by intelligently forwarding alarms with the correct priority and accelerating reaction times.

Discovery

Listening to the Frontline, Competitive Analysis and Conferences - HIMSS

To truly understand the cognitive load nurses face, we couldn't just guess. I researched nursing and spoke with nursing staff to understand their mental models and environmental constraints. I attended HIMSS in Chicago and talked to nurses, vendors and medical professionals. I conducted competitive analysis on nurse call systems.



Key Insight

Nurses don't have time to "read" a dashboard; they need to "scan" it. In critical moments, every second spent interpreting a chart is a second away from a patient.

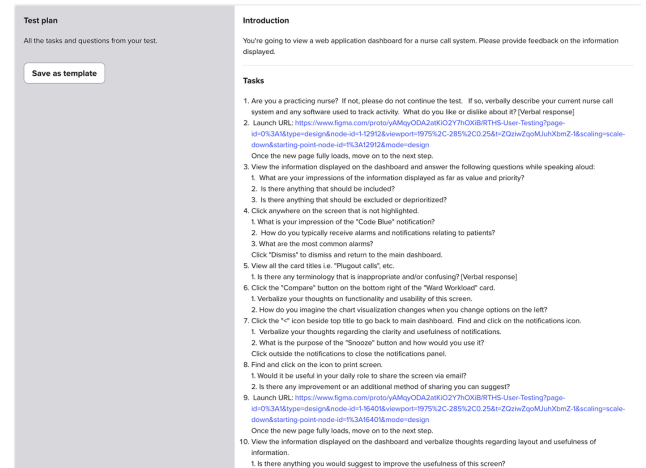
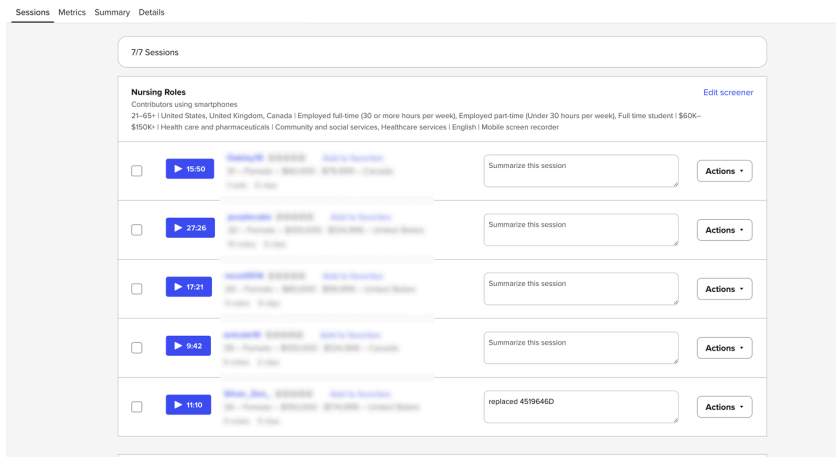
The Takeaway: The interface needed to rely on pre-attentive attributes (like color and size) to communicate urgency instantly.

Validation Iterating with UserTesting.com

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The Feedback Loop: Early feedback from UserTesting revealed that users struggled to differentiate between "Active Calls" and "Waiting Calls" when they were grouped together.

The Pivot: Based on this data, we separated these metrics into distinct visual cards and added trend indicators (e.g., "23% Decrease") to provide immediate context without requiring mental math.



Concepts, Prototypes and Dev Handoff

Nurse Call Analytics (Collaboration with an agency to deliver)

Nurse Call Dashboard that featured customization capabilities as well as metrics about active calls, average response times and occupied beds.

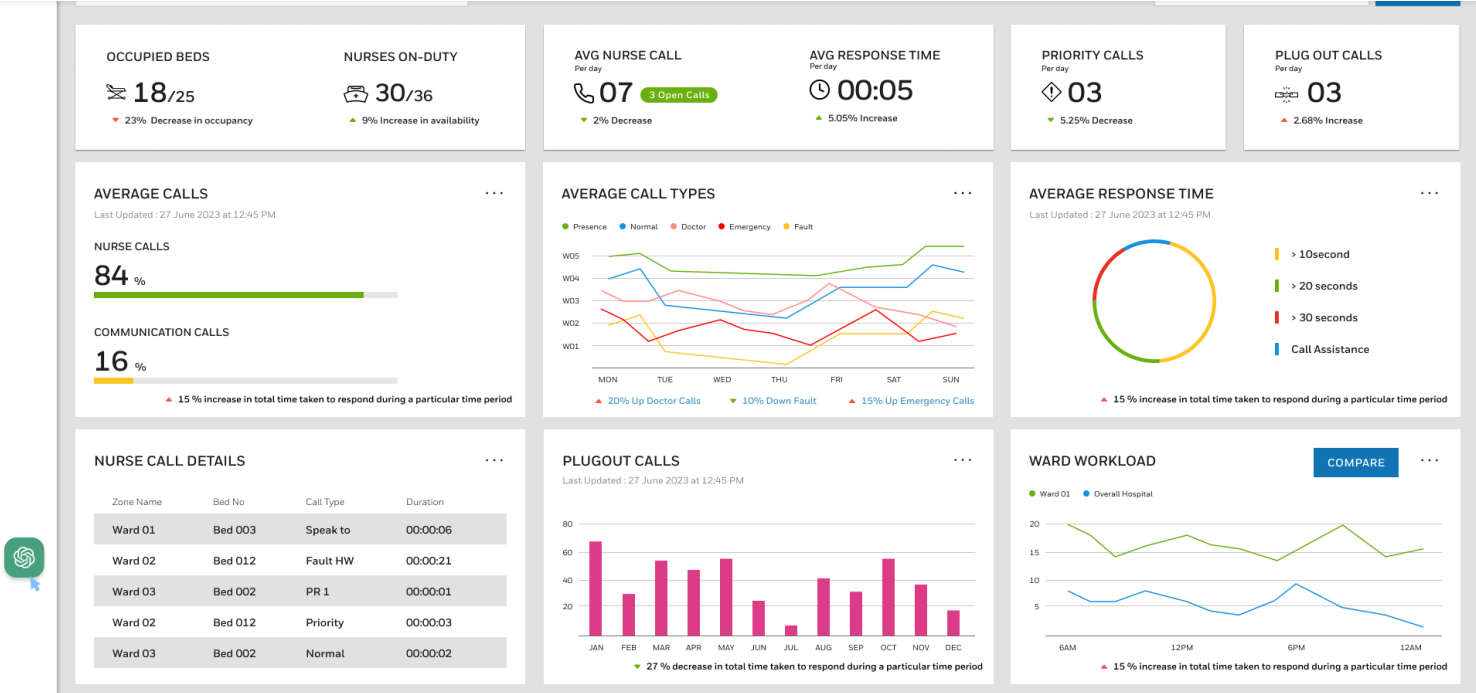
The web dashboard is the central hub for analyzing and managing call activity. It utilizes data visualization to help staff prioritize tasks:

- **Prioritization:** The dashboard tracks Priority Calls (e.g., 3 Open Calls displayed in one view) and calculates the Average Response Time (e.g., 00:05 seconds in one snapshot), showing response data broken down into categories like Immediate, >5 seconds, >10 seconds, and Call Assistance.

- **Call Type Breakdown:** The system displays the percentage of calls categorized as Nurse Calls (84%) versus Communication Calls (16%). It tracks trends for specific events like Emergency Calls (with a reported 15% increase in one visual trend) and Doctor Calls (20% up in one view), displayed across weekdays to help managers allocate resources (W01 to W05).

- **Color-Coded Urgency:** Alerts within the system are clearly categorized and quantified (e.g., Code Blue (42 total, 5% decrease), Code Green (38 total, 18% decrease), Code Red (24 total, 10% increase)

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TOTAL CALLS

📞 112

▲ 10% Increase

PRESENCE CALLS

● 42

▼ 12% Increase

NORMAL NURSE CALLS

● 38

▼ 8% Decrease

DOCTOR CALLS

● 24

▲ 5% Increase

EMERGENCY CALLS

● 05

▼ 10% Decrease

FAULT CALLS

● 23

▲ 2% Increase



All Calls

☒ All Calls

☒ Group 1

☒ Doctor Call

☒ Nurse Call

☒ Vital Call

☒ Group 2

☒ PR 1

☒ PR 2

☒ PR 3

CANCEL

APPLY

CALL TYPE (32)	CATEGORY (5)	DURATION	STATUS	CLOSURE	STAFF	ACTI
Dining Call	Presence	01:20	Closed	June 10	Nurse 1	Sup
Vital Call	Normal	03:00	Closed	June 10	Nurse 2	Info
Diagnostic	Doctor	01:04	Paused	June 10	Nurse 1	Sup
Fire Call	Emergency	01:23	Initiated	NA	Nurse 1	Rais
Plug-out Call	Fault	01:27	Initiated	NA	Nurse 1	Info
Diagnostic	Presence	01:23	Initiated	June 12	Nurse 1	Sup
Plug-out Call	Normal	01:27	Closed	June 12	Nurse 1	Info
Vital Call	Presence	01:23	Initiated	June 12	Nurse 1	Sup
Diagnostic	Normal	01:27	Closed	June 14	Nurse 1	Info
Fire Call	Emergency	01:23	Initiated	NA	Nurse 1	Raise
Diagnostic	Dcotor	01:27	Closed	June 15	Nurse 1	Inform
Vital Call	Doctor	01:23	Initiated	June 15	Nurse 1	Suppress

< 1 2 3 4 5 ... 32 >

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Call Types

Search Call Types

01 Vital Call

02 Doctor Call

03 Nurse Call

04 PR1

05 PR2

06 PR3

07 Dinning Call

08 Staff Call

09 Telephone Call

10 Call

11 Fire Alarm

12 Bathroom/WC Call

13 Diagnostic Call 1

14 Diagnostic Call 2

15 Heart Emergency Call

16 Priority Call

17 Plug-out Call

18 Diagnostic Call 2

Configurations

1

Call Type & Assignment

2

Call Display Configuration

3

Call Special Functions

Generic display configuration

Color

#EE3124

Text Color

#ffffff

Display Symbol

Vital Call

Display Frequency

Perm off

Lamp Frequency

Perm off

Display Symbol Preview

Buzzer Frequency

Perm off

Device Specific Configuration

Select the device and configure



Systevo Corridor Lamp /
Electronic Module configure



Systevo Information Display



Systevo Touch IP /
Care View IP



Systevo Connected Lifecare
Mobile

Systevo Information Display

Color

#EE3124

Text Color

#ffffff

Display Symbol

Vital Call

Text Frequency

Perm off

Buzzer Frequency

Perm off

Display Symbol Preview

CANCEL

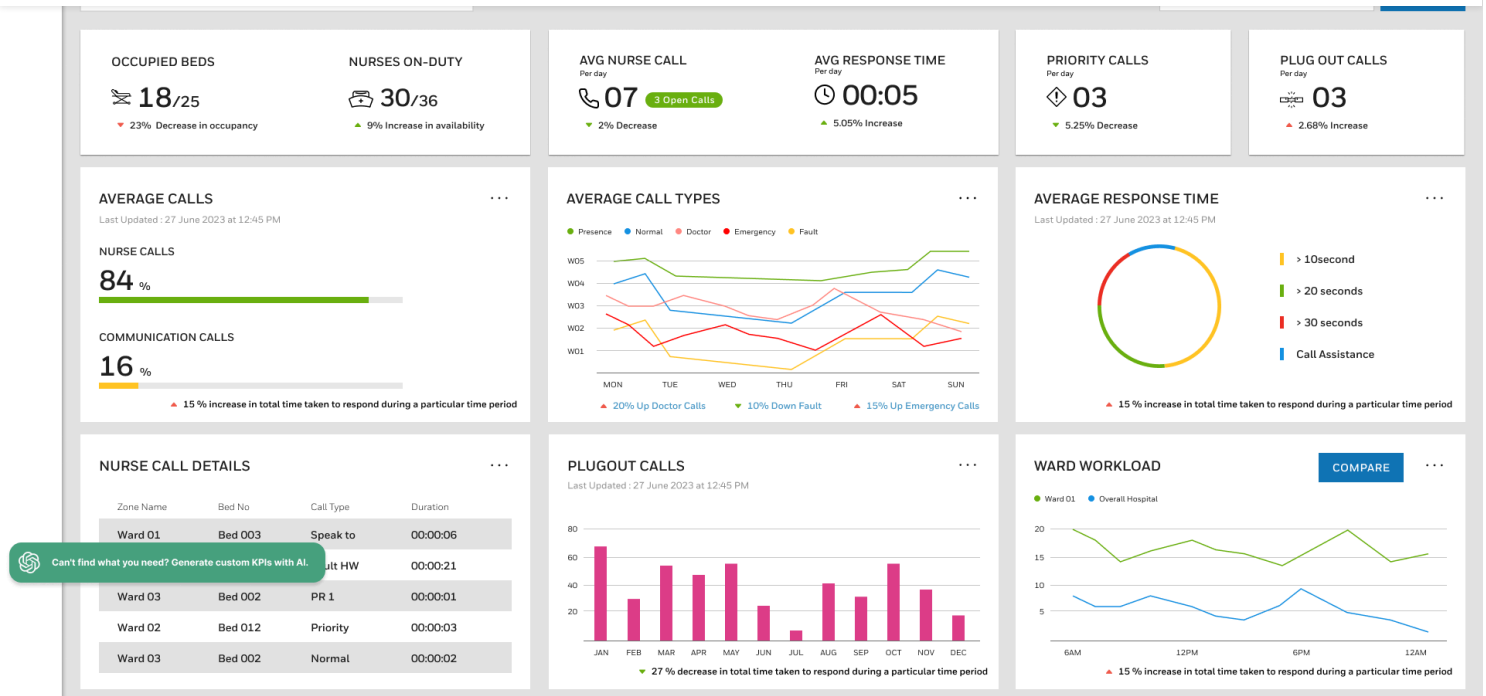
RESET TO DEFAULT

APPLY & SAVE

PREVIOUS

NEXT

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ILC Medication Admin App

Workflow Improvements: The ILC Medication Admin App focused on simplifying the critical task of medication administration, which suffered from a disjointed workflow in the older app. Based on field research insights that favored a predictable flow, the redesigned solution followed a clear linear progression:

- 1. Scan Patient:** The nurse scans the patient's barcode (or enters the ID) using the enterprise mobile device's scan engine. Upon scanning, the patient's profile is loaded, and the nurse can view medication times organized by Morning, Afternoon, Evening, and Scheduled appointments.
- 2. Scan Meds:** The user is then prompted to Start Scanning Medications by pressing buttons on the sides of the mobile device to scan barcodes. As medication is scanned, a toast message confirms the "Medication Found" (e.g., "Triple Antibiotic Ointment") and auto-closes after 1500ms, minimizing distraction. The scanned item moves to the top of the list and is checked off from the planned 5/5 scanned count. Nurses can swipe left to disable a medicine from the scannable list.
- 3. Confirm & End Session:** Once all medications are scanned, the screen automatically navigates to the "Administer Medication" screen, where the nurse confirms administration. The final step, End Session, provides a confirmation popup, distinguishing itself from the confusing "Submit" action of the prior design. The system uses color-coded safety states on the confirmation screen (Green for administered, Red for un-administered).

Validation: User research was crucial to validate the design changes, involving prototype testing with healthcare professionals.

Quantitative Findings (Inferred Task Improvements):

- **Faster Response/Speed:** The design achieved an average scan-to-confirm time improved by 28% in

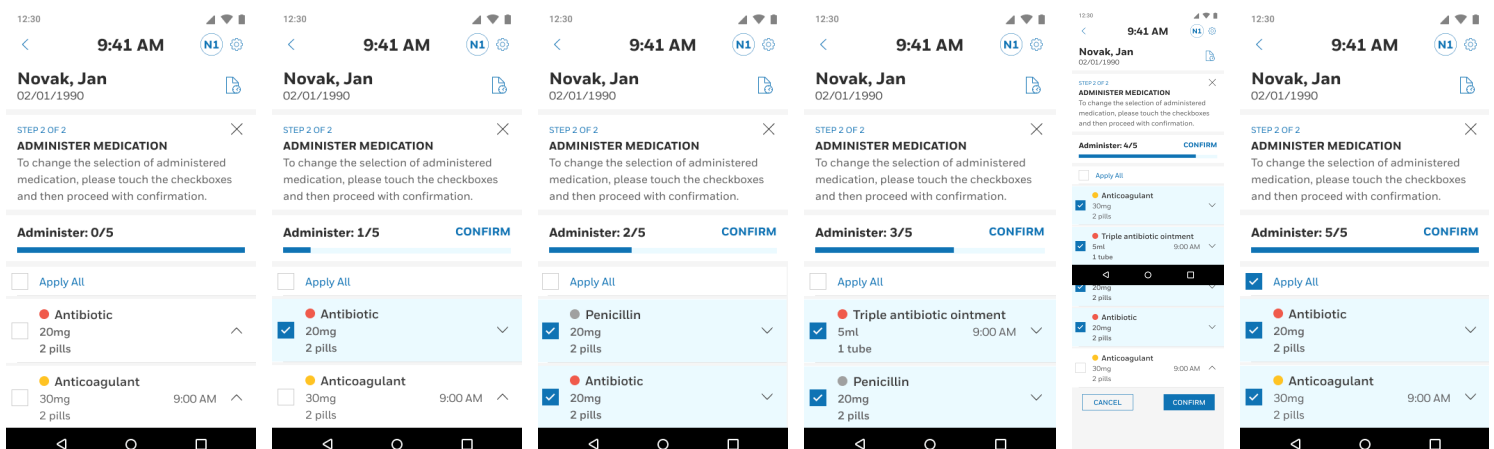
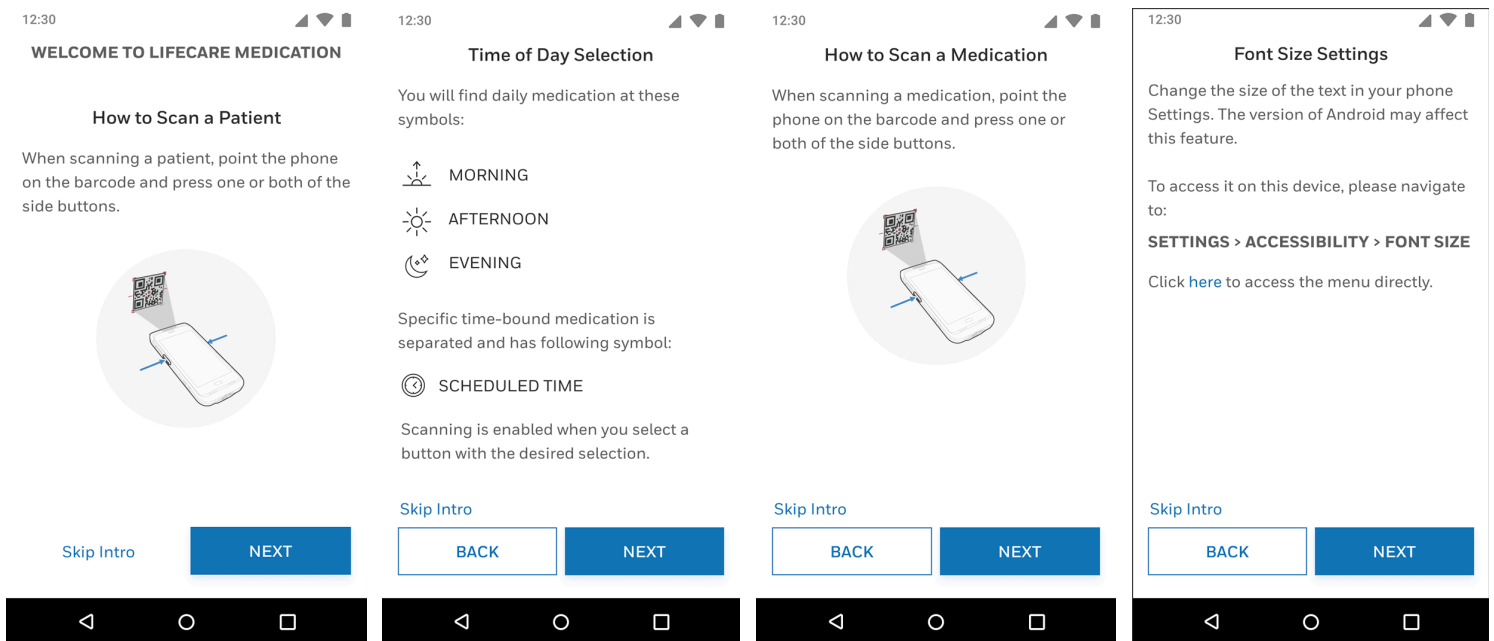
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like a checklist”.

- Reduced Friction: Users appreciated that the automatic closure of toast notifications after scanning reduced screen clutter, noting, “I don’t have to tap to move on—it just works”.
- Increased Confidence: The inclusion of a “View Details” link before final confirmation gave nurses “confidence to verify before proceeding”.
- Accessibility: Users experienced increased satisfaction due to the accessibility prompt linking to system font-size controls, which helped those with visual strain.

4. Impact & Conclusion

The Integrated Lifecare Platform achieved its goal of integrating critical clinical communication (Systevo Nurse Call) and medication administration (ILC Medication App) into seamless, reliable experiences guided by a unified design language. This unification ensured clinical-grade usability that adheres to regulatory standards and aligns with actual nursing cognitive flow, translating real-world urgency into clear, trustworthy flows



Novak, Jan

02/01/1990



STEP 2 OF 2



ADMINISTER MEDICATION

To change the selection of administered medication, please touch the checkboxes and then proceed with confirmation.

Administer: 4/5

CONFIRM

☐

Apply All

● Anticoagulant



30mg
2 pills



CANCEL

CONFIRM



Novak, Jan

02/01/1990



STEP 2 OF 2



ADMINISTER MEDICATION

To change the selection of administered medication, please touch the checkboxes and then proceed with confirmation.

Administer: 0/5

☐

Apply All

● Anticoagulant



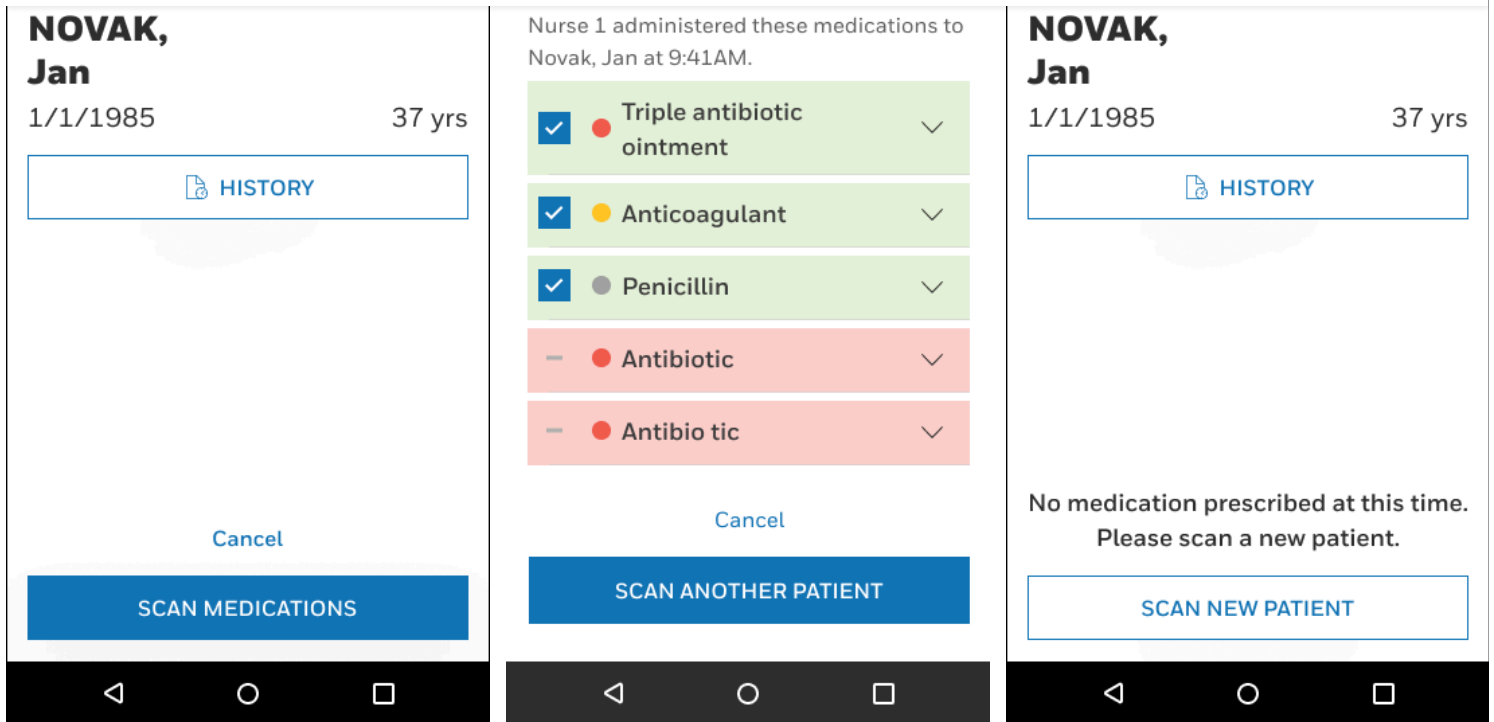
30mg
2 pills



CANCEL



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Systevo Nurse Call Admin Mobile App & QR Code Integration

(Collaboration with design agency)

Add controllers using QR code and step by step guidance

The core mobile solution for nursing staff is the Systevo Connected Life Care mobile application, developed by Honeywell Intelligent Life Care (ILC).

- **Function and Device:** This app is an extension of the traditional nurse call system, allowing caregivers to receive patient calls as push notifications on a work mobile device anytime, anywhere on the move. It is designed to run optimally on specialized Honeywell enterprise mobile devices (e.g., EDA52HC, CT30 XP), which feature a scan engine and special surface treatment for easy disinfection.
- **QR Code Login and Onboarding:** Mobile app users gain easy access by scanning an on-boarding QR-code generated by a system administrator or by using an authorized username and password. The login session is intuitive and user-friendly.
- **Mobile Call Management:** Once logged in, the caregiver can assign the incoming call to herself/himself, which informs other staff that the request is being handled. The mobile app also features a "callback" intercom function for voice communication with the patient, providing quicker assistance when a physical presence in the room is not immediately required.

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Select Work Mode

(Role) Configurations



Commissioning

Lorem ipsum dolor sit amet, consectetur adipisicing elit.



Configuring

Lorem ipsum dolor sit amet, consectetur adipisicing elit.



Operations

Lorem ipsum dolor sit amet, consectetur adipisicing elit.



Diagnostics & Troubleshooting

Lorem ipsum dolor sit amet, consectetur adipisicing elit.

GET STARTED

< Scan Controller



NH Hospitals

📍 HSR Layout, Bangalore

0/4 Tasks completed



Scan Controller QR Code

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

SCAN QR



Projects



Device

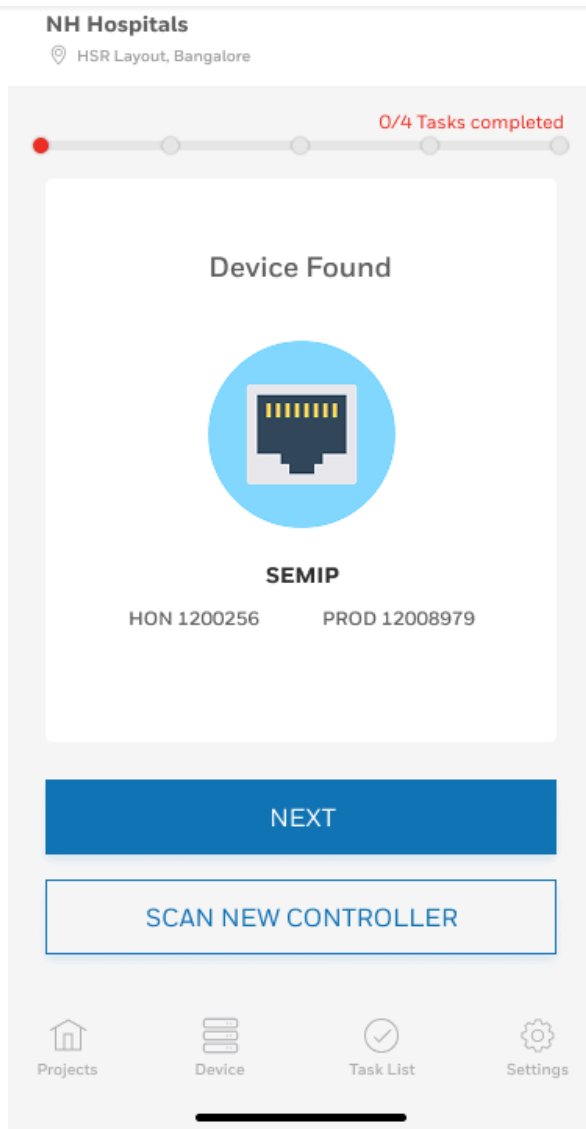


Task List



Settings

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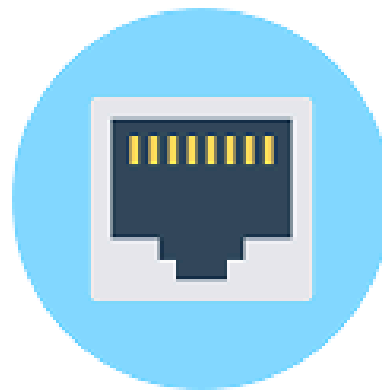
< Scan Controller

NH Hospitals

📍 HSR Layout, Bangalore

0/4 Tasks completed

Device Found



To use this device as a controller,
switch-on the physical button on
the device

NEXT

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SCAN NEW CONTROLLER



Projects



Device



Task List



Settings

< **Controllers**



NH Hospitals

HSR Layout, Bangalore

0/4 Tasks completed



SEMIP



Device is not configured

Controller



Projects



Device



Task List



Settings





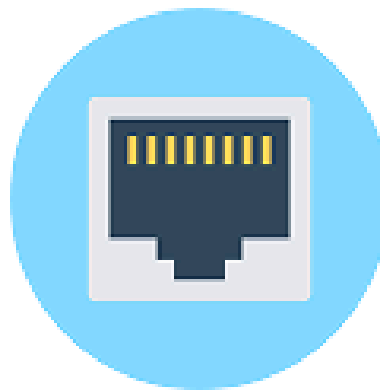
NH Hospitals

 HSR Layout, Bangalore

0/4 Tasks completed



Do you want to configure
this device?



SEMIP

HON 1200256

PROD 12008979



DELETE DEVICE

CONFIGURE DEVICE

CANCEL



Projects



Device



Task List



Settings

Call management administration system on the web

Guided steps for the complex process of add a nurse call system to a hospital.

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Call Types

Search Call Types

01 Vital Call

02 Doctor Call

03 Nurse Call

04 PR1

05 PR2

06 PR3

07 Dinning Call

08 Staff Call

09 Telephone Call

10 Call

11 Fire Alarm

12 Bathroom/WC Call

13 Diagnostic Call 1

14 Diagnostic Call 2

15 Heart Emergency Call

16 Priority Call

17 Plug-Out Call

18 Diagnostic Call 2

19 Heart Emergency Call

20 Priority Call

Configurations

1 Call Type & Assignment

2 Call Display Configuration

3 Call Special Functions

Generic display configuration

Color

#EE3124

Text Color

#ffffff

Display Symbol

Vital Call

Display Frequency

Perm off

Lamp Frequency

Perm off

Buzzer Frequency

Perm off

Display Symbol Preview

Device Specific Configuration

Select the device and configure



Systevo Corridor Lamp /
Electronic Module configure



Systevo Information Display



Systevo Touch IP /
Care View IP



Systevo Connected Lifecare
Mobile

Systevo Corridor Lamp / Systevo Electronic Module

Color

#EE3124

Chamber Selection

1

Lamp Frequency

Perm off

CANCEL

RESET TO DEFAULT

APPLY & SAVE

PREVIOUS

NEXT

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Call Types

Search Call Types

01

Vital Call

02

Doctor Call

03

Nurse Call

04

PR1

05

PR2

06

PR3

07

Dinning Call

08

Staff Call

09

Telephone Call

10

Call

11

Fire Alarm

12

Bathroom/WC Call

13

Diagnostic Call 1

14

Diagnostic Call 2

15

Heart Emergency Call

16

Priority Call

17

Plug-Out Call

18

Diagnostic Call 2

19

Heart Emergency Call

20

Priority Call

21

Diagnostic Call 2

Configurations

1

Call Type & Assignment

2

Call Display Configuration

3

Call Special Functions

Generic display configuration

Color

#EE3124

Text Color

#ffff

Display Symbol

Vital Call

Display Frequency

Perm off

Lamp Frequency

Perm off

Display Symbol Preview

Buzzer Frequency

Perm off

Device Specific Configuration

Select the device and configure

Systeve Corridor Lamp / Electronic Module configure

Systeve Information Display

Systeve Touch IP / Care View IP

Systeve Connected Lifecare Mobile

Systeve Information Display

Color

#EE3124

Text Color

#ffff

Display Symbol

Vital Call

Text Frequency

Perm off

Buzzer Frequency

Perm off

Display Symbol Preview

CANCEL

RESET TO DEFAULT

APPLY & SAVE

PREVIOUS

NEXT

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The screenshot displays a configuration interface for a healthcare system. On the left, a sidebar lists 20 call types, including Vital Call, Doctor Call, Nurse Call, and various Priority Response (PR) calls. The main area is titled 'Configurations' and features a progress bar with three steps: 1. Call Type & Assignment, 2. Call Display Configuration (currently active), and 3. Call Special Functions. The 'Generic display configuration' section includes settings for Color (#EE3124), Text Color (#ffff), Display Frequency (Perm off), Lamp Frequency (Perm off), Buzzer Frequency (Perm off), and Display Symbol (Vital Call). The 'Device Specific Configuration' section allows selecting a device from four options: Systeve Corridor Lamp / Electronic Module configure, Systeve Information Display, Systeve Touch IP / Care View IP, and Systeve Connected Lifecare Mobile. The 'Systeve Touch IP / Systeve Care View IP' section has settings for Color (#EE3124), Buzzer Frequency (Perm off), and Lamp Frequency (Perm off). At the bottom, there are buttons for CANCEL, RESET TO DEFAULT, APPLY & SAVE, PREVIOUS, and NEXT.

Call Types	Configurations
01 Vital Call	1 Call Type & Assignment 2 Call Display Configuration 3 Call Special Functions
02 Doctor Call	Generic display configuration
03 Nurse Call	Color: #EE3124, Text Color: #ffff, Display Symbol: Vital Call
04 PR1	Display Frequency: Perm off, Lamp Frequency: Perm off
05 PR2	Buzzer Frequency: Perm off
06 PR3	Device Specific Configuration
07 Dinning Call	Select the device and configure
08 Staff Call	Systeve Corridor Lamp / Electronic Module configure
09 Telephone Call	Systeve Information Display
10 Call	Systeve Touch IP / Care View IP
11 Fire Alarm	Systeve Connected Lifecare Mobile
12 Bathroom/WC Call	Systeve Touch IP / Systeve Care View IP
13 Diagnostic Call 1	Color: #EE3124, Buzzer Frequency: Perm off, Lamp Frequency: Perm off
14 Diagnostic Call 2	
15 Heart Emergency Call	
16 Priority Call	
17 Plug-Out Call	
18 Diagnostic Call 2	
19 Heart Emergency Call	
20 Priority Call	

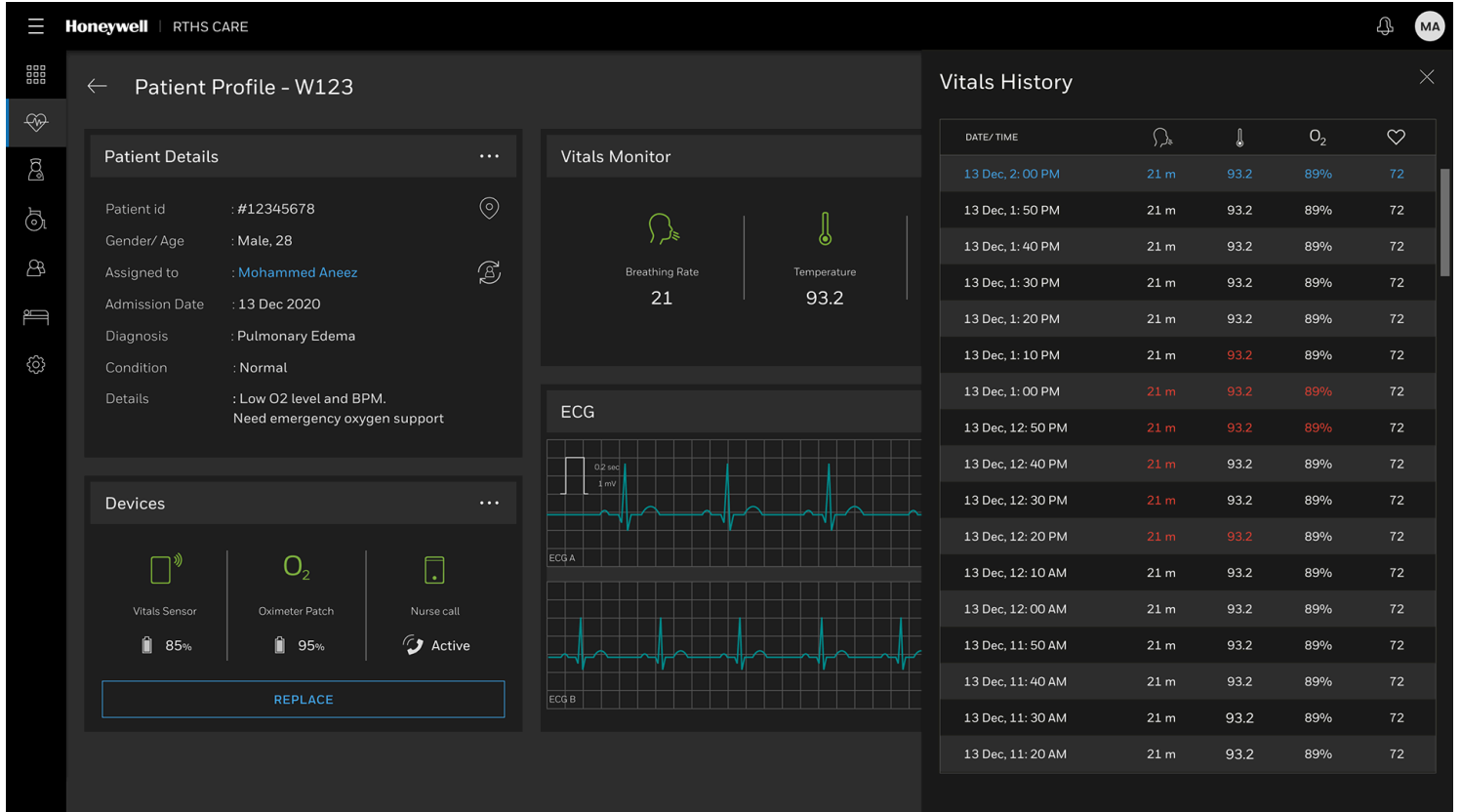
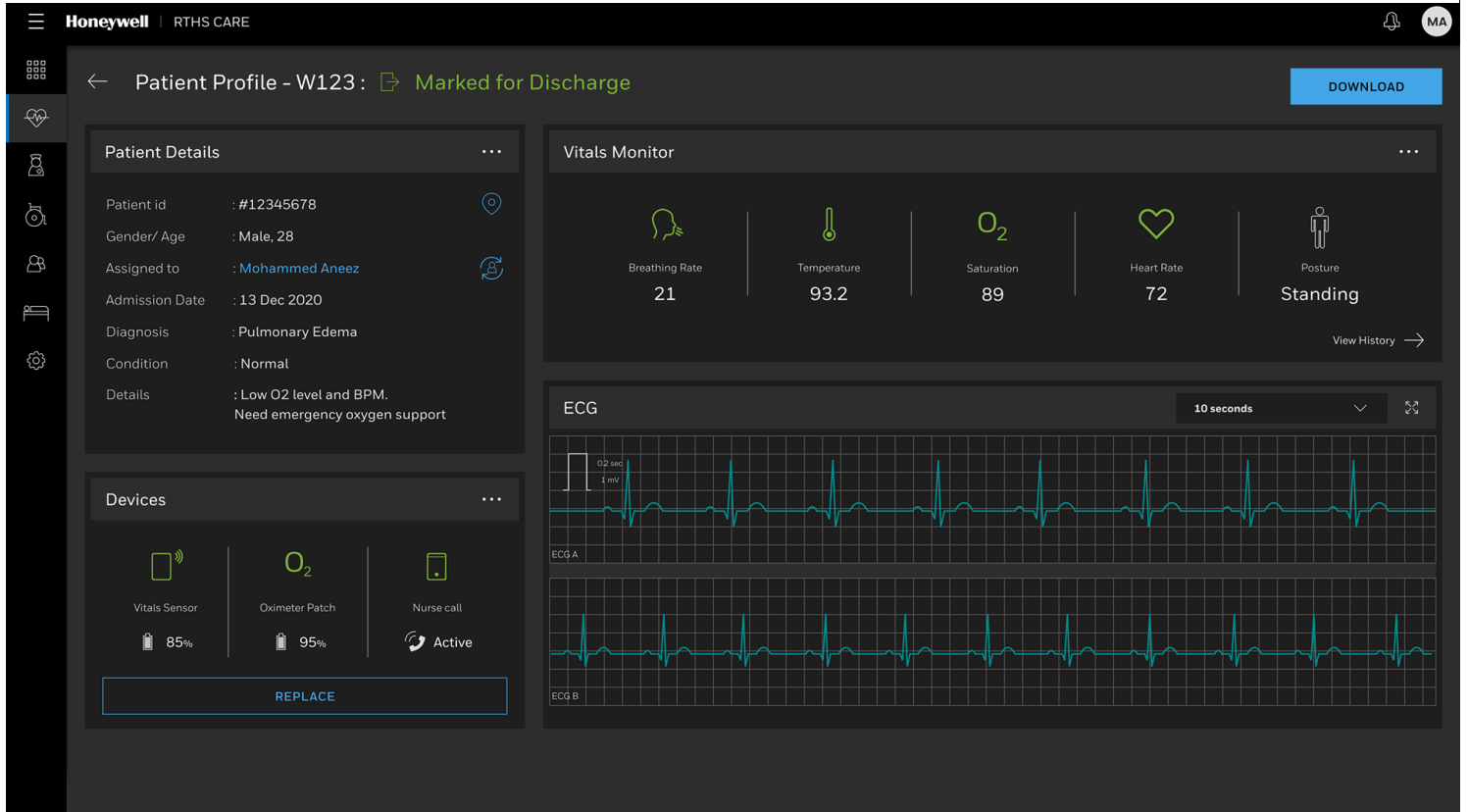
Real Time Health System Dashboard

(Collaboration with Honeywell design team)

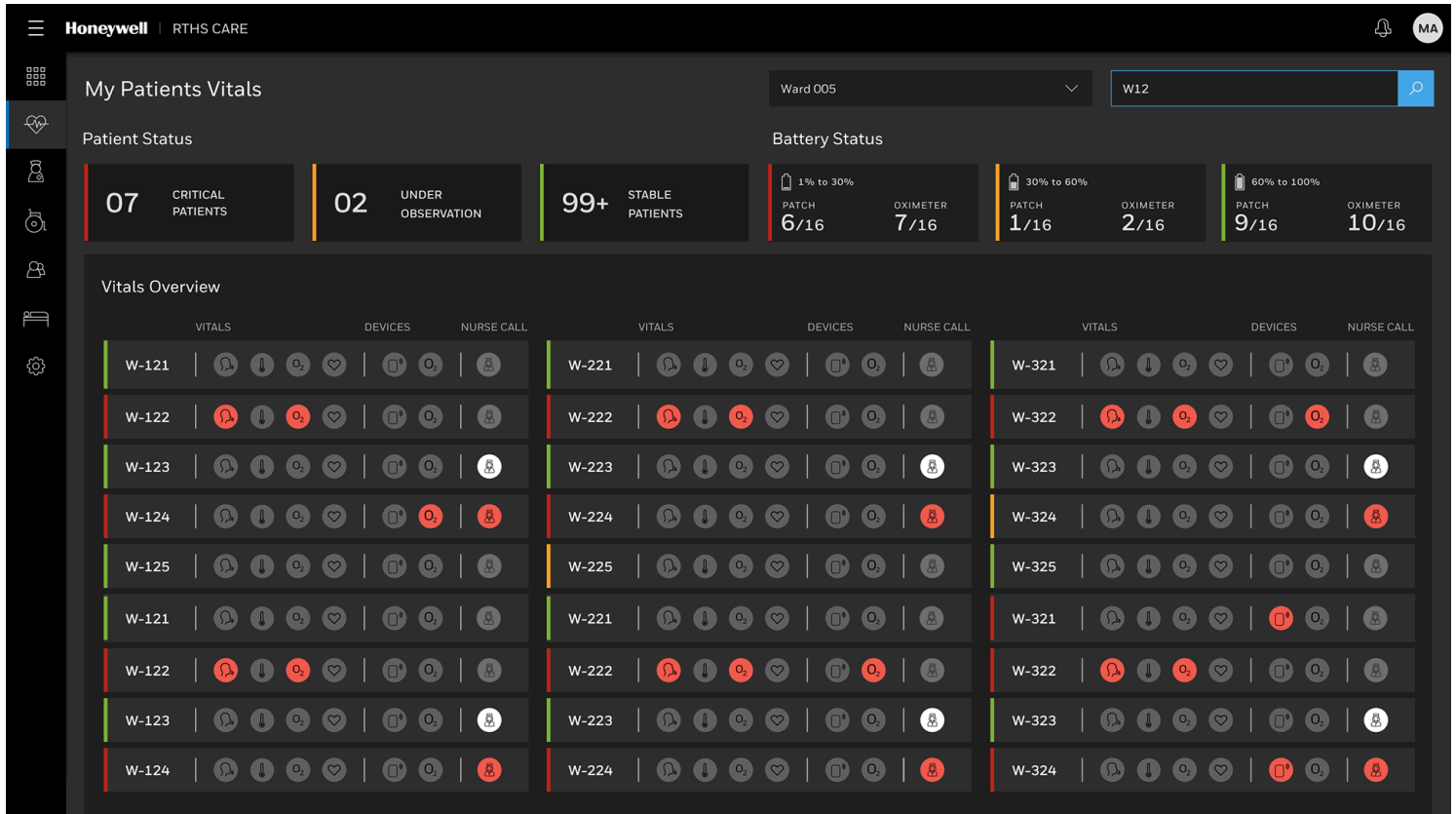
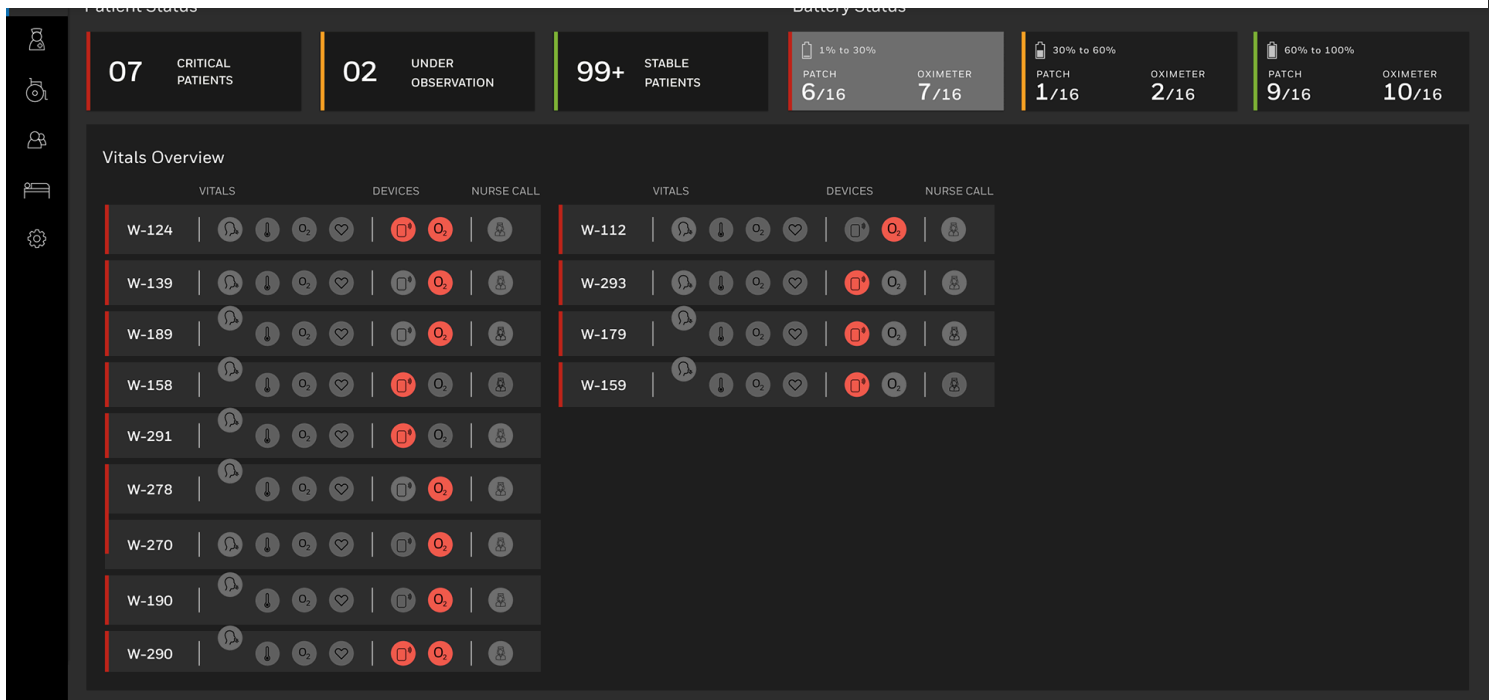
The RTHMS dashboard provides centralized, real-time visualization of patient status, vitals, and bed management information.

- **Real-Time Vitals and Monitoring:** The system continuously monitors patient vitals using a wireless patch with a QR code placed on the patient's chest. Real-time vitals tracked can include ECG, Respiratory rate, Heart rate, Skin temperature, and Posture.
- **Patient Status Dashboard:** The dashboard visualization provides a Vitals Overview that segments patients by severity status, showing counts for Critical Patients, Under Observation, and Stable Patients.
- **Bed Management Display:** The administrative interface includes a Bed Management view that indicates the status of beds, such as Occupied, Bed ready, and Sanitization in process.
- **Alerts and Metrics:** The system utilizes data and analytics processing to generate real-time alerts when clinical thresholds are breached. The dashboard displays operational metrics like the number of Nurses

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TYPE	DESCRIPTION	TIME	ACTION
Replace sensor	Sensor in W 005 has only 5% battery left	01 min ago	<button>REPLACE</button> ×
Replace sensor	Sensor in W 008 has only 3% battery left	02 min ago	<button>REPLACE</button> ×
Low O2	W 007 oxygen level is 76%. Assigned nurse, Shivan is busy with W 009	02 min ago	<button>REPLACE</button> ×
Open call	Open call from W 005, assigned nurse shyam is delayed	03 min ago	<button>RE-ASSIGN</button> ×
Open call	Open call from W 007, assigned nurse shyam is delayed	04 min ago	<button>RE-ASSIGN</button> ×
Discharge	W 004 has been discharged	04 min ago	×
Sensor Replaced	W 012 sensor has been replaced by Nurse John Jack	04 min ago	×
Discharge	W 001 has been discharged	05 min ago	×
High Workload	Nurse Shyam is facing a high workload	06 min ago	<button>RE-ASSIGN</button> ×
Nurse crunch	Ward 4 needs 2 nurses	06 min ago	<button>ASSIGN</button> ×
Open call	Open call from W 012, assigned nurse shyam is delayed	06 min ago	<button>RE-ASSIGN</button> ×
Replace sensor	Sensor in W 018 has only 9% battery left	07 min ago	<button>REPLACE</button> ×
Low O2	W 007 oxygen level is 56%. Assigned nurse, Shivan is busy with W 031	08 min ago	<button>REPLACE</button> ×
Discharge	W 003 has been discharged	09 min ago	×

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